

1. INTRODUCTION

As the amount of online content continues to grow, people are having an increasingly hard time consuming and processing that content at the same rate as it is being produced. For instance, hundreds of thousands of news articles are generated on a daily basis from around the world, and it is no longer a reasonable expectation to rely on readers to sift through lengthy content to find what is important to them. The issue is not the unavailability of information rather it is the overwhelming amount of information available combined with the amount of time and cognitive effort it takes the average person to process that information in timely manner that has created an actual barrier. Thus, this has led researchers towards automated summarization of text, using computational methods to condense a long document into a shorter version that retains the essential meaning of the original document. This is an appealing solution because a good summarization system could support the functionality of a news digest service, now and in the future, act as an assistant in research, provide assistance with legal documents, and provide access to long text for the reading of many individuals. The implications for this technology are not only immediate and practical; they are also more contemporary in nature than academically based.

Over the past few years, advances in deep learning technology have taken over as the prevalent method for generating summaries from textual content, especially following the emergence of transformer models based on pre-trained neural networks. However, there are numerous challenges presented to these models. These models must be able to comprehend actual rather than lexical meanings, retain factual accuracy and keep coherence throughout multiple documents greater in length to their original data, and generate output summaries that appear to be composed in an organic manner. This paper focuses on addressing each of these challenges in news article summarization.

The main focus of this paper is abstractive summarization of English news articles using the CNN/Daily Mail dataset, rather than giving extracts from the original documents as seen in extractive summarization. The study compares four transformer model architectures such as T5 Base, T5 Large, BART CNN Large, and PEGASUS Large. Their performance is evaluated using ROUGE and BLEU, both before and after fine-tuning, to determine which of the models produces the best summary on the dataset.

2. PAPER SUMMARY

The paper primarily provides a comparative analysis and does not present an alternative theoretical approach to summarizing news articles. Comparisons are made among four pre-trained transformer-based models that have all undergone the same procedures for pre-processing and training with respect to each other. They have been evaluated using ROUGE-1, ROUGE-2, ROUGE-L, BLEU, METEOR, before and after fine-tuning. The purpose of these comparisons is to determine how much of the improvement comes from the fine-tuning phase of training. The system utilizes an abstractive summarization approach that is to say, the output generated by those models will contain new sentences rather than simply extracting them from the original article. Additionally, this study also examines whether or not automatic scoring systems are reliable indicators of human judgment by evaluating the correlation between evaluation scores and three characteristics (completeness, correctness and conciseness) of the generated summaries. Finally, statistical analyses are applied to compare the reliability of ROUGE and BLEU.

3. METHODOLOGY ANALYSIS

This section talks about the architecture used in the paper.

3.1. T5-Base and T5-Large

In T5, every NLP task is treated as a text-to-text one, this means that the task is performed entirely in text with no other form of representation involved in the process. In the case of summarization, the input to the

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model is the original article and the output will be the summary. The smaller T5-Base version has approximately 220 million parameters and has a maximum input length of 512 tokens, whereas T5-Large has 770 million parameters, thus allows for much more complex language models to be created using the larger model.

3.2. BART CNN-Large

BART is a model that combines two ideas into one. The encoder reads the whole text in both directions (like BERT) to get the context, and the decoder makes the output one step at a time (like GPT). The CNN-Large version also uses convolution layers to better find local patterns. The fact that BART is trained ahead of time sets it apart. It learns by fixed different kinds of broken text, like sentences that are out of order, words that are missing, and parts that are missing. This helps it understand and put text back together, which is helpful for making summaries. BART has about 139 million parameters and can handle up to 1024 tokens, so it can read longer articles better than T5 models.

3.3. PEGASUS-Large

PEGASUS was made just for abstractive summarization. It is pre-trained in a way that is very similar to the task of summarizing, unlike general models. The main idea behind it is Gap-Sentence Generation. The model takes out whole sentences from a document and learns how to put them back together using the text that is left over. This is a lot like summarizing, which means the model has to know what the content is and give the most important parts. Because of this, PEGASUS is already to summarize before fine-tuning. It has about 568 million parameters and can handle up to 1024 tokens.

3.4. Feature Fusion and Final Prediction

The inference process of all four models is identical. The encoder first gets the article, which makes a contextual representation of the whole text. The decoder then makes the summary step by step, using both the encoder output and words that have already been made. It stops when it gets to an end token or the 128-token limit. The final tokens produced by each of the models will then be converted back into a human-readable summary. Because the models work independently of one another, there are no combinations or ensembles between the four models.

3.5. Dataset(s) used

All experiments were performed using the CNN/Daily Mail dataset, which is a widely utilized benchmark for evaluating news story summarization performance. This dataset consists of more than 300,000 individual articles along with their corresponding human-generated highlight summary called “highlight”. This dataset is divided into three parts, approximately 287,113 for training, 13,368 for validation, and 11,490 for testing, and each news article has an average word count of approximately 692 words and its corresponding highlight summary consists of an average of approximately 52 words. One challenging issue with transformer-based models is that they have a limit on the length of their input sequences. Given that many of the articles are long, truncating articles to process them as input limits how much of each article can be processed by the model. However, both the BART and PEGASUS architectures use a maximum length limit of 1024 tokens, allowing them to process a larger slice of the article without having to truncate it.

3.6. Training Procedure

Transfer learning is used to train the models. All four models use pre-trained versions instead of starting from scratch. They are then fine-tuned on the CNN/Daily Mail dataset. This saves time and resources because it would be very expensive to train such big models from scratch. During fine-tuning, all models will have to create new summaries based on comparing the model’s output summaries against the real summary, then reduce the model’s error (cross-entropy loss) by back-propagating the differences. Also,

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while monitoring the models to measure performance, validation methods have to take place during model training so that the models will not become overfitted. T5 uses AdamW optimizer while BART and PEGASUS use Adam optimizers for model training, will all four models use learning rate schedules with warmup periods. However, the training parameters for models slightly differ, T5-base has a total of 200,000 steps with a batch size of 64, T5-large has 100,000 steps with a batch size of 32. On the other hand, BART and PEGASUS models both have a total of 60,000 steps and a batch size of 64. Moreover, the paper states that to do all four of these models on a low-end Core i3 CPU with low memory (12 GB RAM), implies that either model training should have been limited or other resources was used for the training like a GPU.

3.7. Evaluation Metrics

This study evaluates generated summaries using ROUGE, BLEU and METEOR.

3.8. Baseline Used for Comparison

This paper uses two baseline comparisons. One, the pre-training baseline, indicates how well each model works on the CNN/Daily Mail test set before fine-tuning. This serves as a benchmark for how well the models are able to operate solely through their initial pre-training, and serves to evaluate the contribution to the model's performance resulting from fine-tuning. Another baseline is generated to measure the performance of all models once they have been subjected to fine-tuning, in order to provide a summary of each model's performance respective strengths and weakness relative to its performance across different metrics. So that the comparative result of each of the models can be defined, the study shows that, in general, there will be no single model that outperforms the others on every metric. In addition to comparing against other after fine-tuning, the study also compares the performance of all models against previous studies that provided common baseline for METEOR evaluation, including Lead-3, Pointer-Generator and BART-Large model, without CNN variant.

Baseline performance metrics (pre-training)					
Sl. no	Models	ROUGE-1	ROUGE-2	ROUGE-L	BLEU
1	T5-base	0.29	0.1071	0.2015	0.0431
2	T5-large	0.2429	0.0829	0.1682	0.0338
3	Bart-large CNN	0.1352	0.0628	0.0948	0.0628
4	PEGASUS-large	0.2493	0.0850	0.1664	0.0317

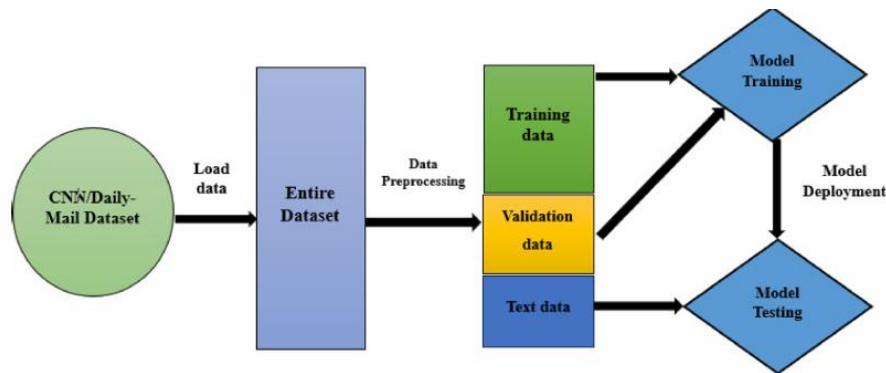


Figure 3: Generating a summary from the input document

4. EXPERIMENTAL SETUP

This section summarizes the results obtained after fine-tuning.

4.1. Results

Final performance (after training)					
Sl. no	Model	ROUGE-1	ROUGE-2	ROUGE-L	BLEU
1	T5-base	0.3279	0.1288	0.2354	0.0565
2	T5-large	0.3414	0.1340	0.2442	0.0575
3	BART CNN large	0.3388	0.1399	0.2458	0.0668
4	PEGASUS large	0.3037	0.1250	0.2173	0.0501

The most apparent finding from the analysis is that all four models experience a dramatic improvement in performance when fine-tuned. Specifically, the BART CNN-Large model initially has the worst pre-training scores (ROUGE-1: 0.1352). After fine-tuning ROUGE-1 score is 0.3388, which represents over a 20% improvement, meaning that the BART pre-trained checkpoint is highly flexible to adapt for a specific task, despite lack of effectiveness at the time of pre-training. T5-Large achieves the highest ROUGE-1 score (0.3414), but only by a small amount over the BART CNN. While T5 has a slight edge in terms of ROUGE-1 performance, the BART CNN outperforms T5 in ROUGE-2 (0.1399), ROUGE-L (0.2458), and BLEU (0.0668) metrics. It is worth noting that ROUGE-2 and ROUGE-L scores are generally regarded as better predictors of the quality of a summary than ROUGE-1, because they represent phrase-level overlap and sequential coherence, rather than only counting words. The fact that BART outperformed T5 on all three metrics forms the basis of our conclusion that BART is the best model overall. The most surprising finding was observed with PEGASUS-Large. It is the most purposefully created model designed for abstractive summarization, yet it performed the lowest of all models.

5. CRITICAL ANALYSIS

After our critical analysis, several limitations remain:

5.1. Limitations

One of the main limitations of the study was input truncation. T5-Base and T5-Large model can only process a maximum of 512 tokens each, whereas the typical length of a CNN/Daily Mail article is longer than that. Because of this, the later portions of the majority of articles must be cut off before a summary can be created by the model. All news articles typically contain their most important information at the beginning, however, losing the rest of the content could potentially have an impact on how well the summary captures the main idea of the article. This limitation arises from the underlying architecture of the model and cannot be addressed simply through better training of that particular model. The BART and PEGASUS models provide up to 1024 tokens, therefore, they typically encounter the issue less than T5 due to their ability to create summaries from longer input documents. Another limitation is that, the summaries are evaluated through ROUGE and BLEU, which only measure word similarity with a reference summary and does not include verifying factual correctness.

5.2. Dataset Bias or Limitations

The dataset has several issues. For instance, the reference summary that accompanied each article on the mobile interface of a news website was written in short bullet-point form, which was not necessarily intended to be the best way to summarize an article. This has resulted in some instances where the summary misses an important part of the article rather than summarizing the article in the best way possible, the model trained to copy the author's specific writing style. Another issue is that the dataset was created using only English language, therefore there is a language bias, topic bias and style bias.

5.3. Reproducibility Concerns

The authors mention the code on Zenodo and its dataset on Kaggle. However, there are several information that were not described well enough. Specifically, the paper does not provide the precise tokenizer used for each model. The reason that these details matter is that there is variability in the results of transformer fine-tuning from one run to another. Without this, it is nearly impossible for others to replicate the results as stated within the paper.

5.4. Scalability Issues

In the study, the reported training environment was a low-powered desktop computer and would make testing T5-Large and PEGASUS-Large very challenging. The paper does not address how it was possible to run the training over T5 or PEGASUS on such low-powered systems. The lack of clarification creates an issue with reproducibility, since others trying to use that same hardware will likely be unable to reproduce the experiments because it has not been specified. Therefore, if the authors used cloud GPUs or have some other resource to support the training, that should be stated in the paper.

6. FUTURE RESEARCH OPPORTUNITIES

The future directions for summarizing medical reports is a potential area. Like newspaper readers, healthcare practitioners also experience an overwhelming amount of information, however, because of the potential negative consequences associated with summarizing, practitioners experience an even greater degree of stress related to making errors in summary generation. Furthermore, truncation problem for long articles is also impactful direction for immediate improvement alongside factual consistency.

7. REFERENCES

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